

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	N.Somasekhar Naidu	Reg. No.:	192472347
Section / Batch:	CSE-AI	Date:	

Code of Conduct

I _N.Somasekhar Naidu-192472347_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- **Gender and number agreement**
- **Recency**
- **Grammatical role**
- **Semantic compatibility**
- **Discourse coherence**

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

- 1. Maintain entity coherence using assignment, mistakes, and you.**
- 2. Use a Cause–Effect or Elaboration discourse relation.**
- 3. Include at least two keywords from:**
 - **review**
 - **improve**
 - **confident**
 - **feedback**
- 4. Response length must be 2–3 sentences.**

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solutio n Develo pment	Applicat ion of Concept s	Justificati on	Sub. Total	Total %
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1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Reference Resolution and Discourse Coherence

Given paragraph: "Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

a) Referring expressions and possible antecedents

Referring Expression	Possible Antecedents
He	Ravi / Arun
him	Ravi / Arun
it	the book on Artificial Intelligence
they	Ravi and Arun
the book	the book on Artificial Intelligence

b) Applying the constraints

Gender and number agreement: "He" and "him" must refer to a singular male person, so both Ravi and Arun are grammatically possible. "They" is plural and therefore refers to Ravi and Arun together.

Recency: Arun is the most recently mentioned male noun before "He", so recency favors Arun. However, recency alone is not sufficient.

Grammatical role and discourse coherence: The first sentence introduces Ravi as the subject. A natural continuation keeps Ravi as the main discourse entity: Ravi was looking for the book, and Arun helped Ravi find it.

Semantic compatibility: "it" must refer to something that can be found, so it refers to the book. "They" refers to both Ravi and Arun because two people can discuss the book together.

c) Constraint table

Referring Expression	Possible Antecedent	Valid / Invalid	Reason
He	Ravi	Valid	Gender/number agree; coherent continuation of Ravi as discourse subject.
He	Arun	Possible but weaker	Gender/number and recency fit, but makes the next sentence less coherent.
him	Ravi	Valid	Arun can naturally help Ravi find the book.
him	Arun	Invalid in intended reading	Would imply Arun helped himself and weakens discourse coherence.
it	Book	Valid	The book is the object that can be found.
it	Ravi / Arun	Invalid	Semantically incompatible with "find it" in this context.
they	Ravi and Arun	Valid	Plural agreement and coherent joint discussion.
the book	AI book	Valid	Direct repeated reference to the previously introduced book.

d) Final coreference chains

- Ravi → He → him → they
- Arun → Arun → they
- Book on Artificial Intelligence → it → the book

e) Paragraph with pronouns replaced

Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.

f) Ambiguity if "He was looking for a book" appeared before introducing Arun

If Arun had not yet been introduced, “He” would most naturally refer to Ravi because Ravi would be the only available singular male antecedent. Introducing Arun before the pronoun creates an additional possible antecedent and therefore increases ambiguity.

Conclusion: Reference resolution should combine agreement, recency, grammatical role, semantic compatibility, and discourse coherence rather than relying on a single constraint.

Q2. Dialog Act Recognition and Response Generation

a) Required dialog acts

- Reassure – reduce the student's worry about possible mistakes.
- Advise – suggest a useful next action concerning the submitted assignment.

b) Important entities to maintain

- you – the student
- assignment – the submitted work
- mistakes – possible errors in the assignment
- review / feedback – actions used to improve the work

c) Three possible responses

Response 1: “It is normal to feel worried after submitting your assignment, but you can still learn from any mistakes. Review the feedback when you receive it so you can improve and feel more confident in your next assignment.”

Response 2: “Do not worry too much about the mistakes in your assignment because they can help you learn. Use the feedback to review your work, improve weak areas, and become more confident.”

Response 3: “You have already submitted the assignment, so focus on what you can learn from the result rather than worrying about mistakes. When feedback is available, review it carefully to improve your work and become more confident.”

d) Comparison of the responses

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
1	High	All constraints satisfied	High	Directly addresses assignment and mistakes
2	High	All constraints satisfied	High	Clear reassurance and advice
3	Very high	All constraints satisfied	Very high	Strong cause-effect flow and practical advice

e) Best response and justification

Response 3 is the best response. It maintains the entities assignment, mistakes, and you; uses a clear cause–effect relationship; includes the keywords feedback, review, improve, and confident; stays within 2–3 sentences; and maintains a positive, polite, and non-contradictory tone.

f) Effect of losing entity coherence

If entity coherence is not maintained, the response may suddenly refer to unrelated work, another person, or a different problem. This makes the conversation confusing and reduces relevance because the user can no longer clearly connect the response to the submitted assignment and possible mistakes.

Conclusion: A good conversational response must satisfy the required dialog acts while maintaining entity continuity, coherence, relevance, and politeness.

Q3. Word Sense Disambiguation and Meaning Representation

Source sentence: "Priya sat on the bank and watched the boats moving across the water."

a) Possible meanings of "bank"

- Bank = a financial institution.
- Bank = the side or edge of a river.

b) Correct sense and contextual justification

The correct sense is the side of a river. The contextual words "boats", "moving", and "water" strongly indicate a river environment. A financial institution is not semantically compatible with watching boats move across water from the bank in the intended sentence.

c) Predicate logic representation

Priya(p)

Bank(b)

RiverBank(b)

Sit(p)

Beside(p, b)

Boat(z)

Water(w)

Watch(p, z)

MoveAcross(z, w)

A compact representation is:

$\text{RiverBank}(b) \wedge \text{SitOn}(\text{Priya}, b) \wedge \text{Boat}(z) \wedge \text{Water}(w) \wedge \text{Watch}(\text{Priya}, z) \wedge \text{MoveAcross}(z, w)$

d) Simple English paraphrase without ambiguity

Priya sat on the side of the river and watched the boats move across the water.

e) How WSD improves Machine Translation

Word Sense Disambiguation selects the intended meaning of an ambiguous source word before translation. Without WSD, "bank" could be translated using the meaning of a financial institution, producing an incorrect sentence. By using contextual clues such as boats and water, the system selects the river-side meaning and can choose the correct target-language word.

Conclusion: WSD improves machine translation by preserving the intended meaning of ambiguous words, reducing lexical translation errors, and producing more coherent translations.